

SARNIA CHRIS HADFIELD, ON, Canada

WMO: 717040

Lat: 42.9950N

Lon: 82.3075W

Elev: 595

StdP: 14.38

Time Zone: -5.00 (NAE)

Period: 05-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	0.7	5.7	-7.6	3.7	1.7	-2.6	4.9	6.4	28.5	22.0	26.0	22.6	9.0	310	0.560

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.8	88.0	72.4	85.3	71.2	82.6	70.0	75.4	84.1	73.7	81.5	72.2	79.3	10.4	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
72.8	124.4	79.9	71.2	117.7	77.8	69.8	111.9	76.6	39.3	84.0	37.7	81.4	36.3	79.5	83.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 24.2	21.2	18.9	DB	-5.2	93.5	7.0	2.4	-10.3	95.2	-14.4	96.6	-18.3	98.0	-23.4	99.7	(4)
(5)			WB	-5.9	79.1	6.8	2.4	-10.8	80.8	-14.8	82.2	-18.6	83.5	-23.5	85.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	48.3	24.9	24.7	33.7	45.0	56.9	65.6	71.0	69.6	63.4	52.3	40.0	30.6		
	DBStd	18.60	10.59	10.75	10.41	8.74	9.14	6.92	5.55	5.16	6.90	8.28	8.75	8.49		
	HDD50	3229	780	708	517	203	31	0	0	0	0	69	318	603		
	HDD65	6661	1244	1127	970	601	286	76	11	15	110	403	751	1067		
	CDD50	2600	1	0	12	52	245	468	651	608	404	142	17	0		
	CDD65	558	0	0	0	0	35	94	197	158	64	10	0	0		
	CDH74	4239	0	0	5	24	380	793	1505	1014	445	73	0	0		
	CDH80	1189	0	0	0	2	99	243	456	254	121	14	0	0		
(14) Wind	WSAvg	9.2	10.4	10.2	10.5	10.7	8.9	8.0	7.1	7.0	7.7	9.4	10.1	10.1		
(15) Precipitation	PrecAvg	34.1	1.9	2.0	2.6	3.0	2.7	3.1	2.7	3.0	3.6	2.5	3.4	2.9		
	PrecMax	43.4	3.3	4.3	5.7	5.2	4.8	5.3	4.3	7.1	9.9	4.7	5.3	4.6		
	PrecMin	25.2	0.9	0.6	1.2	1.5	0.9	0.8	1.2	0.8	0.5	0.7	0.7	1.3		
	PrecStd	4.5	0.8	1.1	1.1	0.9	1.1	1.1	1.0	2.0	2.4	1.1	1.4	0.8		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	53.3	56.9	71.6	77.4	86.6	90.4	92.2	90.1	88.2	81.1	66.9	56.7		
		MCWB	51.7	51.5	61.9	61.8	69.2	72.8	75.3	74.0	73.1	70.2	57.4	54.0		
	2%	DB	47.2	48.9	61.0	71.4	83.0	86.5	88.1	85.8	83.5	74.9	60.8	50.3		
		MCWB	45.2	45.4	53.7	58.6	67.8	70.6	73.3	72.0	71.2	65.0	54.7	47.7		
	5%	DB	42.6	42.8	54.6	65.9	78.4	82.6	85.1	82.7	79.2	69.7	56.8	45.6		
		MCWB	39.9	39.5	49.3	55.3	65.5	69.2	71.7	71.0	69.2	62.2	51.2	42.9		
	10%	DB	38.9	37.8	48.9	60.5	73.4	78.4	81.9	79.6	75.1	65.5	53.0	41.4		
		MCWB	37.0	34.9	44.2	52.1	62.0	67.1	70.2	69.5	66.4	59.0	48.7	38.8		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	52.1	52.6	62.0	63.9	72.9	76.0	78.1	77.7	76.0	71.7	59.2	54.0		
		MCDB	53.2	55.4	71.0	73.0	84.0	86.3	87.5	83.9	85.2	81.0	62.6	56.3		
	2%	WB	45.7	45.6	54.8	60.3	69.8	72.9	75.6	74.7	72.9	67.1	55.5	48.1		
		MCDB	46.9	48.8	58.5	68.3	79.2	81.9	84.5	82.2	80.3	72.0	59.9	50.1		
	5%	WB	40.4	39.2	49.9	56.6	67.0	71.1	73.9	72.8	70.7	63.1	51.9	43.1		
		MCDB	41.9	41.8	54.1	64.7	75.7	79.3	81.4	79.4	76.6	68.3	55.7	45.5		
	10%	WB	37.0	35.3	44.2	52.8	64.1	68.9	72.2	71.2	68.4	59.9	48.7	39.3		
		MCDB	38.8	37.7	48.1	59.3	71.7	75.8	79.2	77.2	73.3	64.8	52.5	41.1		
(35) Mean Daily Temperature Range	5% DB	MDBR	12.3	13.6	14.6	17.5	19.2	17.5	16.8	16.3	17.1	15.1	13.1	10.3		
		MCDBR	14.6	19.5	23.8	28.2	25.6	23.2	21.4	20.4	21.9	21.4	19.5	15.6		
		MCWBR	14.1	17.1	17.9	17.7	14.4	11.9	10.2	10.2	11.3	12.7	14.8	14.6		
	5% WB	MCDBR	14.3	18.6	22.9	24.9	23.2	19.9	18.0	16.7	18.3	18.8	17.5	15.4		
		MCWBR	14.4	16.9	18.5	18.0	14.6	12.0	9.8	9.5	11.2	12.7	14.5	15.0		
(40) Clear-Sky Solar Irradiance	taub		0.309	0.320	0.339	0.382	0.407	0.431	0.418	0.408	0.376	0.359	0.319	0.294		
	taud		2.341	2.303	2.368	2.277	2.263	2.222	2.286	2.302	2.380	2.403	2.485	2.446		
	Ebn at Noon		259	277	286	280	275	268	270	269	270	260	257	257		
	Edh at Noon		26	32	34	40	42	44	41	39	34	29	23	22		
(44) All-Sky Solar Radiation	RadAvg		421	679	1098	1446	1707	1935	1908	1661	1328	841	503	343		
	RadStd		53	73	113	128	146	110	103	77	117	74	66	33		

Historical Trends

		Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (61 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page